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Prevalence and variability in use of physical and chemical restraints in residential aged care facilities: A systematic review and meta-analysis

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ABSTRACT

Background: Use of physical and chemical restraints are common in residential aged care facilities worldwide. Restraint use can pose harm to residents even causing deaths.
Objective: To synthesize the prevalence and variability in physical and chemical restraint use, and examine factors that may contribute to this variability of prevalence rates.
Methods: Six health science databases were searched from inception up to 21st January 2020. Quantitative studies investigating restraint use in residential aged care facilities that reported data from year 2000 onwards were included. Meta-analyses of binomial data using a random effect model were performed to pool proportions of physical or chemical restraints with 95% confidence intervals. Univariable meta-regression analyses were used to assess factors that may contribute to the variability in physical and chemical restraint prevalence. Multiple meta-regression analyses were performed where possible to construct models of factors contributing to these variations.
Results: Eighty-five papers were included. The pooled proportion of physical and chemical restraint use in residential aged care facilities were 33% and 32% respectively. Bedrails (44%) and benzodiazepines (42%) were the most prevalent forms of physical and chemical restraint respectively. Studies from North America (lower prevalence) [coefficient (95% CI): -0.15 (-0.27, -0.03)], measurement approaches using direct observation (higher prevalence) [0.17 (0.02, 0.33)] and a combination of multiple measurement approaches (higher prevalence) [0.17 (0.05, 0.29)] explained 25.5% of variability in the prevalence of physical restraint. Multiple meta-regression analyses were not performed to identify factors that may explain the observed variability in chemical restraint prevalence due to the small number of studies with data available.
Conclusion: Variability in prevalence of physical restraint could be explained partly by different measurement approaches and geographical regions. Valid and reliable measurement approaches across different regions is required to understand cultural differences due to geographical region effects on the prevalence of physical restraint use.

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What is already known about the topic?

- Physical and chemical restraints are commonly used in residential aged care facilities despite little data on their efficacy and benefits.
- Previous studies have reported a large discrepancy in the prevalence of physical and chemical restraints.
- Variance in restraint use between countries and regions may be attributable to factors related to the types of restraint considered and how these have been measured but they are unclear.

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What this paper adds

- The prevalence of physical and chemical restraint use in residential aged care facilities in the last two decades were 33% and 32% respectively. Bedrails (44%) and benzodiazepines (42%) were the most prevalent forms of physical and chemical restraint respectively.
- Measurement approaches using direct observation and a combination of measurement approaches yielded higher prevalence and studies from North America yielded lower prevalence rates, these factors explained 25.5% of variability of physical restraint prevalence.
- Valid and reliable measurement approaches across different regions are required to understand cultural differences due to geographical region effects in the prevalence of physical restraint use.

1. Introduction

The quality of care for older people in residential aged care facilities has received increasing attention in recent decades. This particularly relates to the use of physical and chemical restraints in residents with cognitive impairment who may no longer have capacity to provide informed consent. An international consensus statement defines physical restraints as "any action or procedure that prevents a person's free body movement to a position of choice and/or normal access to his/her body by the use of any method, attached or adjacent to a person's body that he/she cannot control or remove easily." (Bleijlevens et al., 2016). While there has not been an international consensus definition for chemical restraint developed, some studies have referred to chemical restraint as prescribing of psychotropic medications to subdue resident activity, manage behavioural symptoms of dementia or to control behaviour (Gruneir et al., 2008; Lam et al., 2017). Physical and chemical restraints are commonly used despite little data on their efficacy and benefits. Even when the intent to use a restraint is for the safety of the resident, there can be unintended outcomes. Complications of physical restraints can be serious, for example using bedrails or belts to prevent a resident from falling may result in death (Berzlanovich et al., 2012). Older residents may be susceptible to antipsychotic, benzodiazepine and antidepressant related adverse drug events, including falls, excess sedation, stroke and death (Peisah and Skladzien 2014). Psychotropic medications have been linked to lower resident self-reported quality of life (Juola et al., 2016).

It is important to know the prevalence of restraints because of their continued use despite the potential for harm to residents. Previous studies have reported a wide range in the prevalence of physical and chemical restraints. For example, the prevalence of physical restraints has been reported to vary between 26% (Meyer et al., 2009) to 56% (Huizing et al., 2007) in European nursing homes. There are discrepancies between these studies and the types of physical restraints that have been measured. For example, some studies have included and others excluded bedrails as a form of physical restraint (Evans et al., 2002).

Knowledge of prevalence of restraint use is necessary for benchmarking between different areas or facilities and for the evaluation of policies that may impact upon restraint use. This review aimed to synthesize the prevalence of and variability in any physical and chemical restraint use in residential aged care facilities (aim 1). However, to do this we recognized five important factors (Boyle 1998) that may influence the prevalence of and variability in any restraint use. These are: i) prevalence in restraint use may vary between regions around the world due to variations in common care practices, ii) definitions of restraint may vary between studies, iii) the methods for how restraint is measured may vary between studies, iv) the case mix of residents may vary between studies and influence restraint use, and v) the period of time over which restraint use was observed or recalled within resident may vary between studies. Thus, this research also aimed to investigate the associations between each of these factors and the reported prevalence of restraint in each study (aim 2), so that our overall estimates of the prevalence of physical and chemical restraint could be appropriately contextualised. Lastly, we examined the specific mechanisms, equipment and medication used to implement restraint to describe the prevalence of different types of physical and chemical restraints that have been used in residential aged care facilities (aim 3).

2. Methods

2.1. Design

This study was a systematic review and meta-analysis. It was registered with Prospero (CRD 42020176726).

2.2. Search strategy

A database search using Ovid Medline, PsychINFO, Embase, Emcare, Cochrane Central Register of Controlled Trials, and CINAHL Plus was conducted from database inception up to 21st January 2020. The databases were searched from all dates but selection of studies was restricted to the time frame of restraint data reported rather than the publication year. Medical Subject Heading (MeSH) keywords were used in the search strategy based on a modified PICO (Participant-older adults, Intervention-physical, chemical restraint, Context-residential aged care facilities, Outcome-prevalence of restraint) model for Ovid databases, with subject headings considered for the CINAHL database (supplementary material A). Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines were followed for the literature searching process. In addition, the reference lists of the included articles were hand-searched for relevant studies. A grey literature search was not conducted for this review.

A study was selected for inclusion if it: (a) published outcome data of the amount of physical, chemical or other forms of restraint use in residential aged care facilities including nursing homes, assisted living facilities, residential care, care homes or long-term care wards within these facilities, (b) was a quantitative study with research design e.g. randomised controlled trails (RCT), clustered randomised controlled trials, quasi-experimental studies, pre and post-test, cohort studies, longitudinal studies, cross sectional studies or surveys, and (c) focussed on older adults or reported a participant mean age of at least 65 years.

A study was excluded if it was (a) published in languages other than English, (b) a publication that reported data of restraint use before the year 2000, (c) a letter, newspaper article, magazine article, commentary, legal paper, ethics paper, systematic review, case study, qualitative paper, poster abstract or dissertation, (d) a study that reported amount of restraint use on a mixed population of residential aged care residents and hospital inpatients for which data for residents cannot be separated from the aggregate data, (e) a study that described the prevalence of psychotropic or other medications without establishing there was a prescriber intent to restrain or to reduce restraints. In other words, we had only included studies where the study authors had stated that medications were being used for "restraint" purposes or described medications in the context of "restraint" reduction, regardless of whether a definition of chemical restraint was used.

2.3. Outcomes

The primary outcomes extracted from each study where available were:

(i) Prevalence of "any physical restraints" regardless of means (expressed as the proportion of residents restrained by any physical mechanism)
(ii) Prevalence of "any chemical restraints" regardless of means (expressed as the proportion of residents who received any medication for restraint purposes)

The secondary outcome extracted from each study where available was:

Prevalence of specific mechanisms/equipment/medication use to implement restraint (expressed as the proportion of residents restrained by that mechanism e.g. belts, benzodiazepines administered for the purpose of restraint where data for that individual mechanism or medication for restraint was presented).

2.4. Study screening and data extraction

From the search results, two authors (DCAL, LMR) screened all abstracts to identify potentially relevant papers for the review. Full texts of the papers were then independently reviewed to identify papers that were eligible to be included. Any disagreement was resolved by a consensus of the two authors and a third author (TPH) was involved where necessary.

Data were then extracted by two authors (DCAL, LMR) using a data extraction form. These data included study, sample, restraint and measurement characteristics, e.g. study year, country, clinical conditions of the overall sample and subgroups, definition of restraints, the number of sample restrained, sample size, restraint type, measurement approaches and measurement time frame. Any disagreements were resolved through discussion and consensus of an additional author (TPH).

2.5. Risk of bias assessment

We performed an assessment of "risk of bias" across the included studies that were used for meta-analyses. Risk of bias was assessed using the JBI (Joanna Briggs Institute) Critical Appraisal Checklist for Studies Reporting Prevalence Data (Munn et al., 2015). Studies were rated against each of the nine questions in the checklist. Each question can be rated as a "Yes", "No", "Unclear" or "Not applicable", and were related to sampling, data analysis, validity of the methodology, reliability of measurement approach and response rate.

3. Data synthesis

Meta-analyses of binomial data were performed to pool proportions wherever possible using a random effect model. Data for proportions were the number of residents (from the overall sample) restrained by the various mechanisms as the numerator with the total number of residents (overall sample) observed as the denominator. These data were extracted from the baseline data reported in the studies regardless of their research designs. For example, the pre-test data was taken from pre and post-test designs, and the initial data was taken from cohort or longitudinal studies. For RCTs, clustered RCTs and quasi-experimental studies, the number of residents restrained and the number of observations in the intervention and control groups prior to the intervention were added. Subgroup data of residents were not included in the synthesis where studies had reported data for the overall sample. For example, a study that reported restraint use in the overall sample but also stratified the sample into those with or without perineal dermatitis. In this example and studies similar to this, the data for the overall sample was used. For studies that had only reported subgroup data, the number of residents restrained in the subgroups and the number of residents in the subgroups were added to derive the data for the overall sample where possible. A score method was used to calculate study-specific 95% confidence intervals (CI) for the pooled proportion (Nyaga et al., 2014). All studies were pooled in terms of the primary and secondary outcomes where data were available regardless of their country, case-mix, measurement time, measurement approaches, or if any definition of restraints was used. Care was taken not to include studies with the same participants twice in each meta-analysis.

The consistency between studies reporting the primary outcomes of "any physical restraints" and "any chemical restraints" were assessed using I^2 where a value of 0% indicates no heterogeneity and larger values show increasing heterogeneity, and Cochran's Q statistic (Higgins et al., 2003). An I^2 value greater than 50%, i.e. more than 50% of the variation is due to heterogeneity rather than chance is considered substantial heterogeneity (Schroll et al., 2011). For meta-analyses with substantial heterogeneity, five factors were chosen a priori because they are important determinants of prevalence (Boyle 1998) to understand their contribution to the heterogeneity in the reported prevalence rates through meta-regression analyses. Univariable meta-regression analyses were conducted to examine the associations for each of these factors with the primary outcomes of prevalence of "any physical restraints" and "any chemical restraints". These were: a) geographical regions of study (North America, Europe, Asia, Australia and countries from "mixed regions"), b) case-mix of residents (being dementia-specific or not dementia-specific), c) presence or absence of a definition of restraint in the research report (presence of an explicit definition of restraint or an implicit definition describing restraints used for the measurement of outcome, or a combination of these versus none), d) time period used to estimate the period prevalence (this ranged from less than one day to 180 days for physical restraints and 7 days to 30 days for chemical restraints), and e) measurement approach (being direct observation, nursing staff or carers interview, review of entries in local facility records, extraction from administrative datasets, a combination of any of these approaches, or unspecified/unclear measurement approaches).

This univariable meta-regression analysis was followed by construction of a multiple meta-regression model to determine the factors that were independently associated with the heterogeneity in the primary outcomes. Each model was constructed using a backwards elimination approach where dummy variables for each level of each factor were entered into the model to begin with (Hosmer et al., 2013). Independent variables were removed sequentially based upon having the highest adjusted p-value in the model until all independent variables had a p-value <0.05. Each eliminated variable was then re-entered into the model to see if it met the $p<0.05$ threshold with the reduced set of retained predictor variables as covariates. This process was undertaken for both the "any physical restraints" outcome and the "any chemical restraints" outcome. The possibility of collinearity in the predictor (independent) variables entered in the multiple meta-regression model was investigated by computing the tetrachoric pairwise correlations between dummy variables generated from these predictor variables.

Standard error (SE) of the sample proportion used in the univariable and multiple meta-regression analyses was calculated using the mathematical formula SEp = sqrt [p (1 - p) / n] where p is proportion of residents restrained in the sample and n is number of observations in the sample.

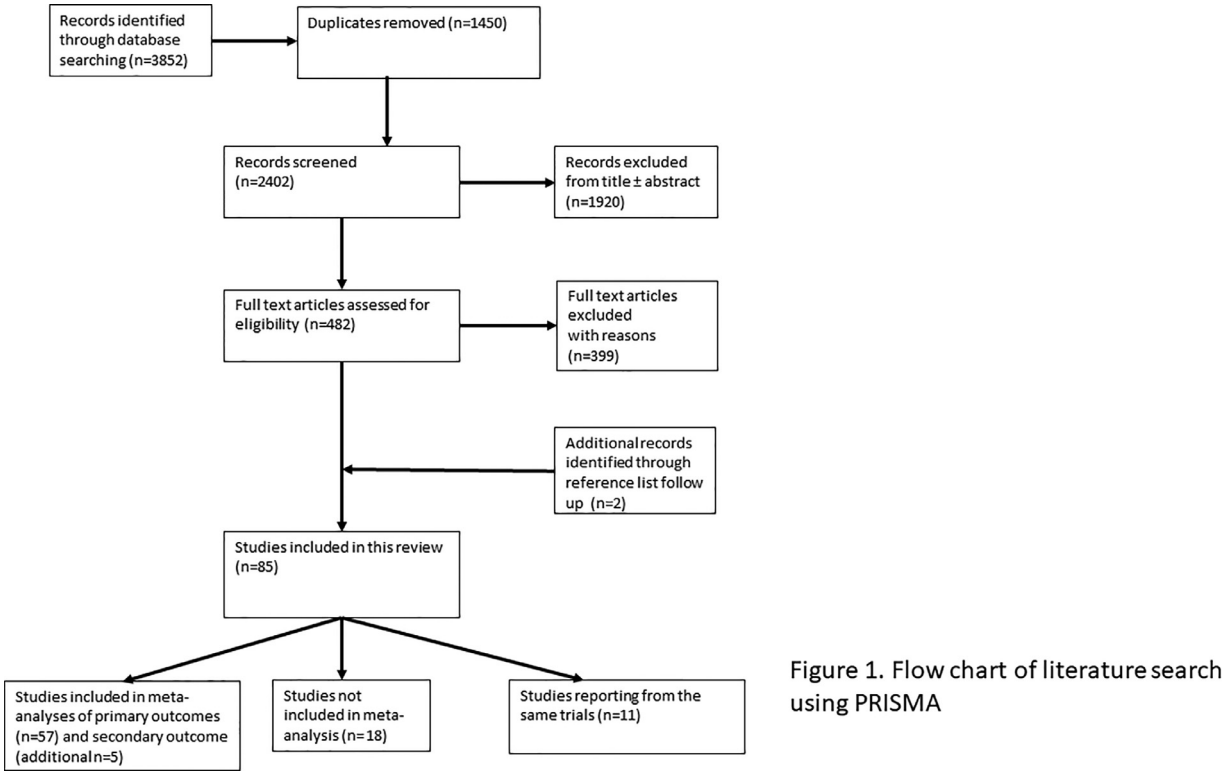

Fig. 1. Flow chart of literature search using PRISMA.

STATA (version 15.1, college station TX) was used for all statistical analyses. The alpha criterion for statistical testing was set at $p<0.05$.

Results for studies that measured restraints under different parameters or using methods other than proportions that were not able to be pooled in the meta-analyses were described separately.

4. Results

There were 85 papers included in this review. Of the papers, 57 were used in the meta-analyses of the primary outcomes and an additional five were used for the meta-analysis of the secondary outcome, and 18 contained data that were unable to be used in meta-analyses. A further 11 papers reported data from the same trails (see Fig. 1). Of these, Gulpers et al. (2013) and Gulpers et al. (2011); Huizing et al. (2007) and two papers by Huizing et al. (2009); Oye et al. (2017) and Testad et al. (2016); Voyer et al. (2005) and Voyer, Verreault, Mengue et al. (2005); Yeh (2003) and Yeh (2002) reported from the same study. The geographical regions of the study country were North America [USA (n = 19 studies), Canada (n = 12)], Europe [The Netherlands (n = 14), Germany (n = 8), Norway (n = 5), Spain (n = 4), Sweden (n = 2), Italy (n = 2), Romania, Switzerland, Finland, Austria, multiple European countries (each n = 1)], Asia [China (n = 6), Taiwan (n = 3), Singapore (n = 1)], Australia (n = 3), and countries from "mixed regions" [European and middle Eastern countries (n = 1)]. Characteristics of the included studies and outcomes examined are presented in the supplementary material Table 1. A list of citations of the included studies is provided in the supplementary material B.

Risk of bias assessment of the studies that were used in the meta-analyses showed that the main issues were related to a majority of studies that had poor or unclear sampling frame appropriate to address the target population, and unclear validity and reliability of data collection methods for restraint use (see supplementary material Table 2).

4.1. Primary outcomes

4.1.1. Prevalence of "any physical restraints"

Data were pooled from 50 studies that reported prevalence of any physical restraints regardless of mechanism. The pooled proportion was 33% [0.33, 95% CI (0.31, 0.35), I^2=99.9%, $p<0.001$] (Fig. 2). There was a substantial heterogeneity between these study results.

Characteristics of the five factors described in the studies that were considered in the univariable meta-regression analyses are presented in Table 1. Measurement approaches and geographical regions of study country were found to contribute to the heterogeneity identified. Specifically, measurement approaches of physical restraints using direct observation (r^2=8.6%) and extraction of data from administrative datasets (r^2=10.7%) yielded prevalence rates that were significantly different from other measurement approaches. In addition, studies from Australia (r^2=6.3%) and North America (r^2=12.3%) had prevalence of physical restraints that were significantly different from other geographical regions (Table 1).

Meta-analyses with data separated into subgroups illustrate the magnitude of difference between different geographical regions. The pooled proportion of any physical restraints was shown to be the lowest in the category "mixed regions" 20% [0.20, 95% CI (0.16, 0.23)], followed by North America 22% [0.22, 95% CI (0.19, 0.24)], 37% in Europe [0.37, 95% CI (0.31, 0.43)], 44% in Asia [0.44, 95% CI (0.34, 0.55)] and highest in Australia 65% [0.65, 95% CI (0.56, 0.73)] (Fig. 3). In addition, measurement approaches using direct observation yielded the highest pooled proportion of physical restraints 49% [0.49, 95% CI (0.32, 0.65)], followed by using a combination of multiple measurement approaches 42% [0.42, 95% CI (0.31, 0.53)], review of data from local facility records 28% [0.28, 95% CI (0.18, 0.38)] yielded similar results as nurses or carers interview 27% [0.27, 95% CI (0.20, 0.35)] while data extraction from administrative data sets yielded the lowest pooled proportion of physical restraints 20% [0.20, 95% CI (0.17, 0.23)] (Fig. 4). Studies

Table 1
Univariable associations between factors considered (independent variables) that may contribute to the heterogeneity between studies and proportion of any physical restraints/any chemical restraints (dependent variable) and their multivariable associations

Factors considered (independent variable)	Characteristics of factors in the studies	Univariable meta-regression Coefficient (95% CI) p value	Multiple meta-regression (final model) Coefficient (95% CI) p value
Proportion of any physical restraints (dependent variable)			
Time period to estimate period prevalence of restraint use[a]	Time range <1 to 180 days (*n*=35 studies)	0.00(-0.00, 0.00) *p*=0.47	N/A
	Unspecified measurement time (*n*=15 studies)		
Absence or presence of any definitions of restraints[b]	No definitions of restraints (*n*=4 studies)	0.01 (-0.21, 0.24) *p*=0.90	N/A
	Some sort of definition used (*n*=46 studies)		
Case-mix of residents[c]	Data for dementia specific population reported (*n*=12 studies)	-0.01 (-0.15, 0.14) *p*=0.94	N/A
	Non-dementia specific population reported (*n*=38 studies)		
Measurement approaches of restraints[d]	Direct observations (*n*=9 studies)	0.17 (0.02, 0.33) *p*=0.03*	0.17 (0.02, 0.33) *p*=0.03*
	Extraction of data from administrative datasets (*n*=11 studies)	-0.18 (-0.32, -0.04) *p*=0.01*	N/A
	Review entries in local facility records (*n*=5 studies)	-0.07 (-0.27, 0.14) *p*=0.51	N/A
	Nurses or carers interview (*n*=7 studies)	-0.08 (-0.25, 0.10) *p*=0.39	N/A
	A combination of multiple measurement approaches (*n*=14 studies)	0.10 (-0.03, 0.24) *p*=0.13	0.17 (0.05, 0.29) *p*=0.01*
	Unspecified or unclear measurement approaches (*n*=4 studies)	-0.00 (-0.23, 0.23) *p*=0.99	N/A
Geographical regions of study country[e]	Europe (*n*=23 studies)	0.05 (-0.07, 0.17) *p*=0.40	N/A
	North America (*n*=16 studies)	-0.17 (-0.29, -0.05) *p*=0.01*	-0.15 (-0.27, -0.03) *p*=0.02*
	Asia (*n*=8 studies)	0.12 (-0.05, 0.28) *p*=0.17	N/A
	Australia (*n*=2 studies)	0.33 (0.01, 0.64) *p*=0.04*	N/A
	Mixed regions (*n*=1)	-0.15 (-0.59, 0.29) *p*=0.49	N/A
Proportion of any chemical restraints (dependent variable)			
Time period to estimate period prevalence of restraint use	Time ranged from 7 days to 30 days (*n*=5 studies)	0.00 (-0.03, 0.03) *p*=0.82	N/A
	Unspecified measurement time (*n*=2 studies)		
Absence or presence of any definitions of restraints	No definitions of restraint (*n*=2 studies)	-0.01 (-0.48, 0.47) *p*=0.97	N/A
	Some sort of definition used (*n*=5 studies)		
Case-mix of residents	Data for dementia specific population reported (*n*=3 studies)	0.32 (0.14, 0.50) *p*=0.01*	N/A
	Non-dementia specific population reported (*n*=4 studies)		
Measurement approaches of restraints	Direct observation (*n*=0 study)		N/A
	Extraction of data from administrative datasets (*n*=4 studies)	-0.11 (-0.54, 0.32) *p*=0.53	N/A
	Review entries in local facility records (*n*=0 study)		N/A
	Nurses or carers interview (*n*=0 study)		N/A
	A combination of multiple measurement approaches (*n*=1 study)	0.20 (-0.37, 0.77) *p*=0.40	N/A
	Unspecified or unclear measurement approaches (*n*=2 studies)	0.01 (-0.47, 0.48) *p*=0.97	N/A
Geographical regions of study country	Europe (*n*=2 studies)	0.13 (-0.31, 0.57) *p*=0.48	N/A
	North America (*n*=3 studies)	0.18 (-0.24, 0.60) *p*=0.31	N/A
	Asia (*n*=2 studies)	-0.30 (-0.56, -0.05) *p*=0.03*	N/A

* Statistical significance at $p<0.05$Coding was applied for each factor described in the studies as follows:
[a] Time period to estimate period prevalence was used as a continuous variable.
[b] Presence or absence of definition was coded as 1 if some sort of definition of restraint were used (for example, an explicit definition of restraint, an implicit definition describing types of restraints used for measuring the outcome, or a combination of these) or 0 if nothing was used in the study.
[c] Case-mix of residents was coded as 0 for studies reporting restraint data for dementia specific or 1 for non-dementia specific populations.
[d] Measurement approaches of restraint use was coded categorically to the data collection methods and included any unspecified/unclear method.
[e] Geographical regions of study countries were coded categorically.

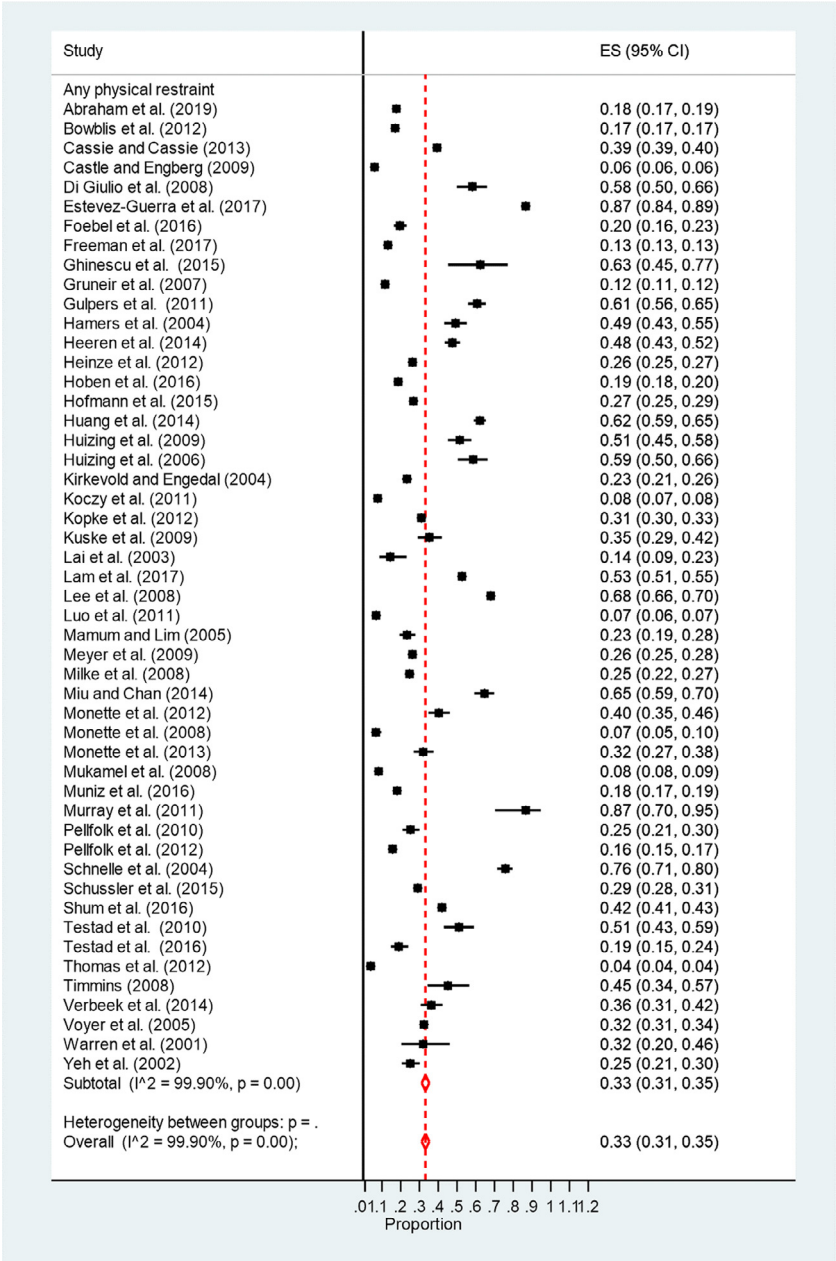

Fig. 2. Pooled proportion of any physical restraints.

with unspecified or unclear measurement approaches in this review had a pooled prevalence estimate of 34% [0.34, 95% CI (0.24, 0.45)].

The variables retained in the multiple meta-regression model were dummy variables coding for studies from North America (lower prevalence) [coefficient (95% CI): −0.15 (−0.27, −0.03)], and using measurement approaches of direct observation (higher prevalence) [0.17 (0.02, 0.33)], and combination of multiple measurement approaches (higher prevalence) [0.17 (0.05, 0.29)] where the outcome was any physical restraints regardless of mechanism. These factors explained 25.5% of the variability in this outcome (Table 1).

Significant pairwise correlations between these variable sets included: North America / data extraction from administrative data sets (rho=0.60, $p<0.01$), Europe / data extraction from administrative data sets (rho=−0.59, $p = 0.01$), North America / direct observations (rho=−1, $p = 0.01$), Europe / direct observations (rho=0.55, $p = 0.03$), Europe / nurses or carers interview (rho=0.63, $p = 0.01$), and Europe / combination of multiple measurement approaches (rho=−0.58, $p<0.01$).

4.1.2. Prevalence of "any chemical restraints"

Data were pooled from seven studies that reported prevalence of any chemical restraint regardless of medication. The pooled proportion was 32% [0.32, 95% CI (0.23, 0.42), I^2=99.9%, $p<0.001$] (Fig. 5). There was a substantial heterogeneity between the studies.

Characteristics of the five factors described in the studies that were considered in the meta-regression analyses are presented in Table 1. Univariable meta-regression identified case-mix of residents (being dementia-specific vs non-dementia specific population) and studies from Asia significantly explained

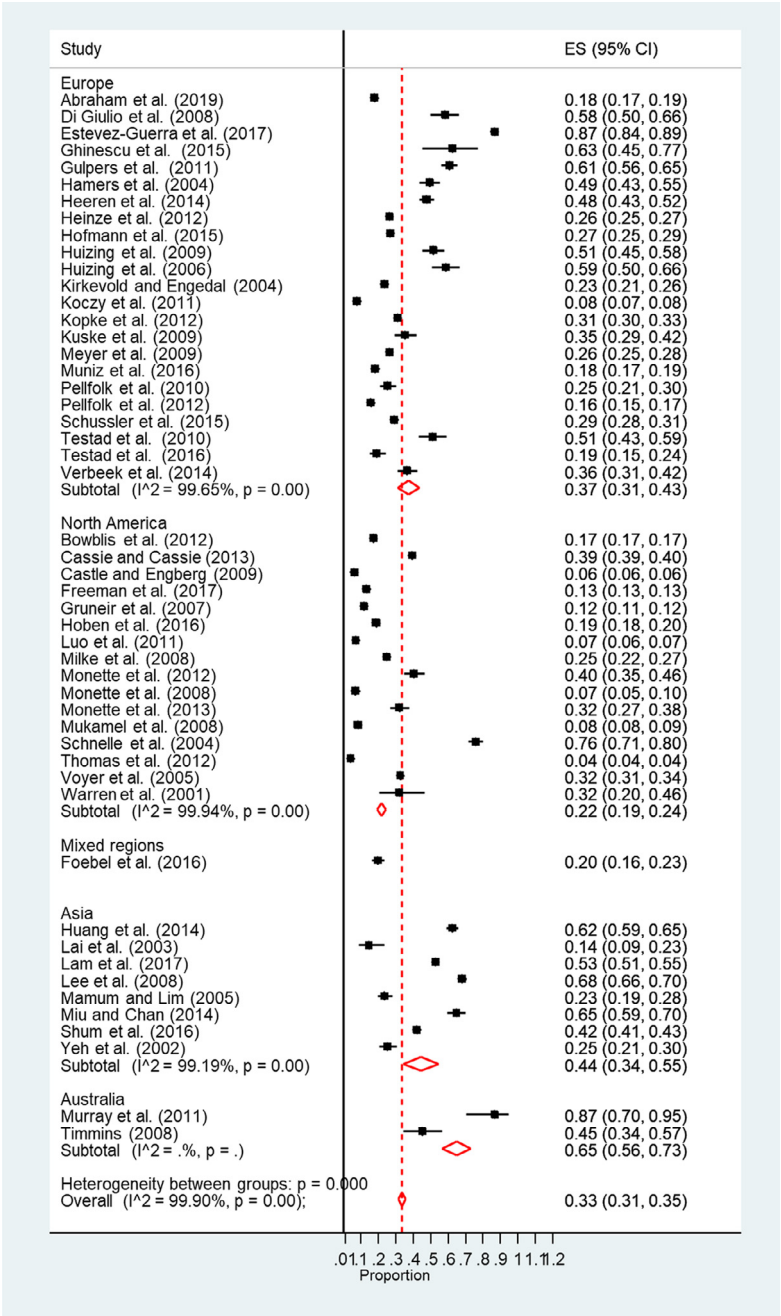

Fig. 3. Subgroup analysis of pooled proportion of any physical restraints by geographical regions.

the observed heterogeneity (Table 1). We did not pursue a multiple meta-regression for this outcome to avoid developing an over-fitted model due to the small number of studies with data available.

4.2. Secondary outcome

4.2.1. Prevalence of specific mechanism /equipment/medication use to implement restraint

Of the 18 types of physical restraints (including environmental or alternative restraints and constraints) that were described in the studies, their pooled prevalence were: bedrails (unilateral or bilateral) 44%; force/pressure used in medical treatment or activities of daily living 14%; chair belts 8%; chair restraints 8%; surveillance/sensors/tracking systems 8%; trunk restraints 7%; "other physical restraints" 7%; "any belt restraint" 6%; chairs or wheelchairs with locked tables 5%; sleep suits 5%; chairs to prevent rising (geri chairs, deep or overturned chairs, chairs on a board) 5%; bed sheet restraints 4%; bed belts 3%; limb restraints (including mittens) 3%; locked bedroom doors 2%; physical retention 2%, removal of walking aids from residents 1% and bedrail protector 1%.

Of the seven types of medications used for chemical restraints, their pooled prevalence were: benzodiazepines 42%, antipsychotics 38%; antidepressants 37%, neuroleptics 29%; antiepileptics 19%; anxiolytics 13% and hypnotics 1%. The studies, pooled proportion and 95% CI for each type of physical and chemical restraints are presented in the supplementary material Table 3.

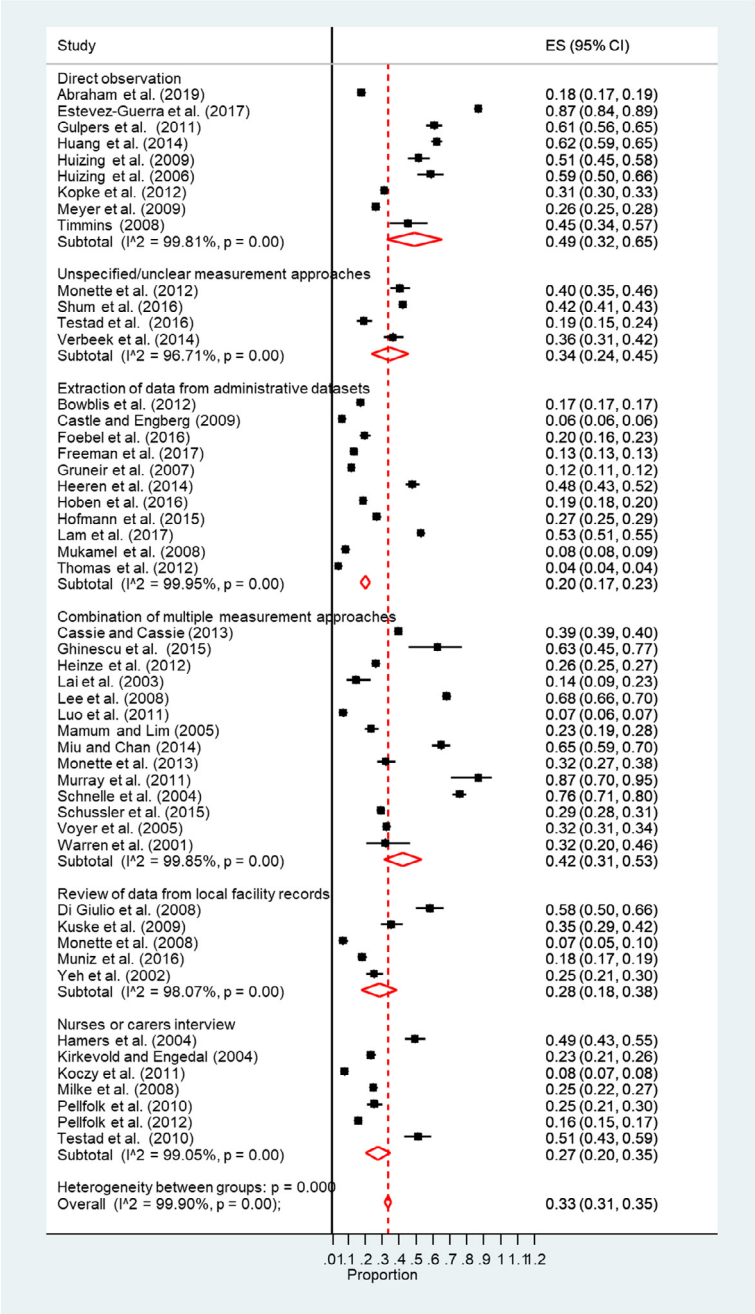

Fig. 4. Subgroup analysis of pooled proportion of any physical restraints by measurement approaches.

4.3. Other measurements of physical or chemical restraint use

There were 18 studies that reported physical or chemical restraint use for which their prevalence estimates could not be pooled in the meta-analyses.

Eight studies reported the prevalence of restraints using a country-level or facility-level data rather than a resident-level data. From the studies in North America, Kim et al. (2013) reported approximately 1% mean prevalence of any physical restraints amongst residents with untreated depression in the USA from 2005 to 2007 (Kim et al., 2013), Dudevich et al. (2018) reported a reduction of physical and chemical restraint use in three Canadian states comparing the data from 2011/2012 to 2015/2016 data (Dudevich et al., 2018), Feng et al. (2009) reported a range of mean prevalence of physical restraint use across North American, Asian and European countries ranging from 9% in the USA to 31% in Canada (Feng et al., 2009), Konetzka et al. (2014) reported a decrease of physical restraint use in the USA from 2000 to 2008 in residents with different severity of cognitive impairment but with a corresponding increase of chemical restraint use (Konetzka et al., 2014), Gordon et al. (2016) reported 2.7% of residents being physically restrained in 33 facilities (Gordon et al., 2016) while Decker and Castle (2011) reported a weighted mean prevalence of 7.4% in facilities with an administrator model versus 7.1% in facilities with a Director of Nursing model in the United States (Decker and Castle, 2011). A European study, Halfens et al. (2013) reported the prevalence of bedrail use was 58% across facilities in The Netherlands and 78% across facilities in Austria (Halfens et al., 2013) while an Australian study, Jeon et al. (2019) reported a 0% median prevalence

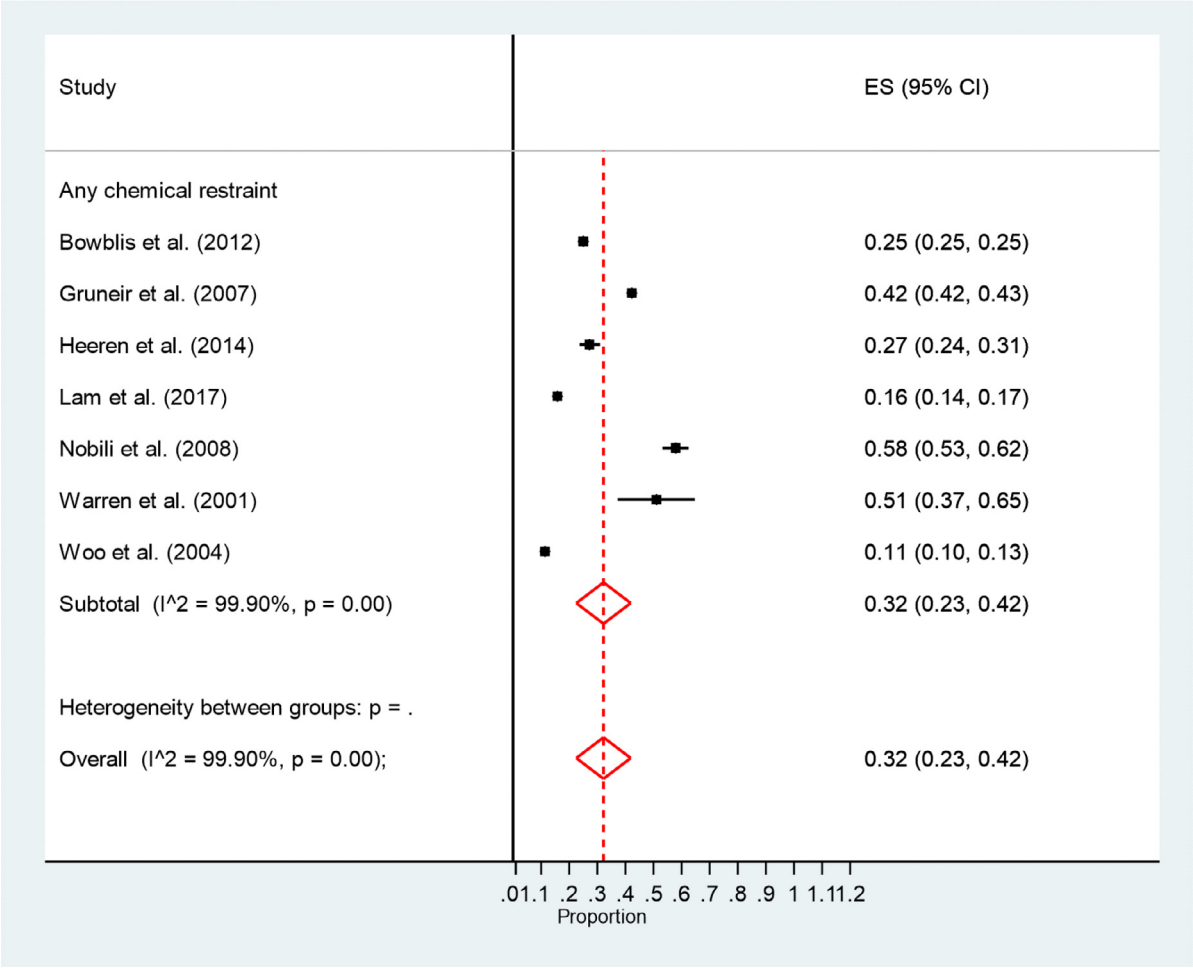

Fig. 5. Pooled proportion of any chemical restraints.

of physical restraint use among its nursing homes with an interquartile range (quartile 1 and quartile 3) of 0 to 7.27 (Jeon et al., 2019).

Five studies reported restraints use under different parameters. Aranda-Gallardo et al. (2018) reported 45% of fall cases had bedrails fitted and 12% had associated use of physical restraints (Aranda-Gallardo et al., 2018), Te Boekhorst et al. (2013) reported 52% of residents had "fixation" and 48% had "restrictive chair" use but these were measured out of a sample of residents with 100% physical restraint use (Te Boekhorst et al., 2013), while Willemse et al. (2016) reported a mean range for different types of physical restraints across multiple facilities (for example 0 to 35% for belt use) and chemical restraints (for example 0 to 67% for antipsychotics) (Willemse et al., 2016). Dever Fitzgerald et al. (2016) reported a mean number of residents restrained, ranging from 3.64 to 4.15 in 148 residents using "nurses reported restraint use" at baseline compared to a mean number of residents restrained, ranging from 2.41 to 2.67 in 98 residents using "observed restraint use" at baseline (Dever Fitzgerald et al., 2016). Joyce et al. (2018) reported a mean of 15.1% of inappropriate antipsychotic use and 7% of physical restraint use in over 5 million admissions based on two estimation (linear probability and instrumental variable) models (Joyce et al., 2018).

Two studies did not report prevalence but instead reported other measurements of amount of restraint use. Allen et al. (2005) reported their residents were being restrained in bed for a mean duration of 312 min among the non-survivors at their end of life (Allen et al., 2005), and 162 min among the survivors when restraints for both groups were measured over one day, while Koczy et al. (2011) reported their residents were being restrained for a mean of 1725 min at baseline for the intervention and control groups measured over three days (Koczy et al., 2011).

Three studies reported an average of number of restraints used. Chang et al. (2013) reported an average of two physical restraints being used for their residents (Chang et al., 2013) while Testad et al. (2005) reported an average of 3.3 physical restraints. However, both were measured over an unspecified period of time (Testad et al., 2005). Warren et al. (2001) reported an average number of 0.6 chemical restraint used in residents with dementia in residential care centres compared to 0.8 in residents with dementia that lived in special care units over a two-week period (Warren et al., 2001).

5. Discussion

Our meta-analyses found a prevalence for use of any physical or chemical restraint being present in 33% and 32% of residents respectively. This estimate is considered reasonable for physical restraints as a large number of studies and many with large samples were used to generate the pooled proportion. However, both physical and chemical restraint prevalence rates need to be treated with some caution due to the large heterogeneity in the estimates and data collection methodologies described between studies. This review identified that the estimate for the prevalence of physical

restraint was impacted by the method used to collect data and the geographic region of the study location. This is demonstrated by the factors that remained in the final multiple meta-regression model of "any physical restraints". Measurement approaches that used a direct observation approach or a combination of multiple measurement approaches yielded higher frequencies indicating that other data collection approaches are likely to under-estimate the true prevalence rate of restraint. These findings indicate that there may be bias in our overall estimate of physical restraint prevalence due to how these data were collected in a majority of studies.

Studies that reported data from the geographical region of North America were associated with a lower level of any physical restraint use. This was even after taking into account the differences attributable to the data collection approach. This indicates that there may be policy settings, practices or differences in the resident population in North America that lead to the lower reported rates of restraint. The variability in physical restraint prevalence between geographical regions were also noted within regions. For example, in Australia where a pooled proportion from two studies demonstrated 65% of residents had physical restraints (Timmins 2008; Murray 2011) versus a single study in which the data could not be pooled (Jeon et al., 2019) reported 0% median prevalence rate among facilities that used physical restraints. These variabilities may have been due to differences in measurement approaches employed by studies between these regions and within these regions. It has been observed in other fields that variation in instruments/surveys used, operational definitions of outcomes, sampling methods and prevalence estimation methods can affect the reliability and validity of estimates obtained (Boyle 1998).

It is also possible that cultural differences in restraint use across these regions exist. Culture includes "everything that is socially developed and learned" (Hufford 1995). In the context of restraint use, the emphasis and use of laws and / or policy to restrict restraint use and the principles that underpin them will be influenced by the particular cultural lens applied. For example, there are numerous factors which may be culturally driven including values related to authority, control and their place in the therapeutic environment, assessment of harm and risks, values applied to the balancing of benefits versus harm, the perceived problems posed by "troublesome" residents, and of the solutions that can be used (O'Connor 1998). In addition, cultural differences can be distinguished in the manifestation of ageism among health care personnel including attitudes, views and expectations applied to older people and caregivers (Wyman et al., 2018). These cultural differences may be seen in the context of laws or policy implementation to restrict restraint use. For example, the data highlights that in North America, the USA has implemented federal laws, the Omnibus Budget Reconciliation Act of 1987 to mandate major changes for health care in nursing homes (Centres for Medicare and Medicaid Services 2008) whereas European countries such as Germany have adopted a least restraint policy as the standard of care (Abraham et al., 2019), Britain has produced guidelines (Bell 1997) to reduce the use of restraints, Asian countries such as Japan have advocated for physical restraint-free care through efforts of individual leaders in the health care industry (Akamine 2000), while Australia adopted aged care standards to minimise the use of restrictive practices including physical and chemical restraints (Aged Care Quality and Safety Commission 2020).

The inclusion or exclusion of certain equipment / mechanisms as physical restraints could impact the overall prevalence estimate of "any physical restraints" or that of the individual restraint equipment/mechanism. For example, the use of bedrails was not assessed in several studies but bedrails have been found as the most prevalent physical restraint. Even though some included studies have specified the definition of physical restraints as "any method / physical/ mechanical/ materials/ equipment….which restricts freedom of movement or normal access to one's body" (Lai et al., 2003; Mamun and Lim 2005; Freeman et al., 2017), environmental or alternative restraints such as surveillance / sensors / tracking systems or constraints such as use of physical force / pressure in medical treatments or activities of daily living are noted to be less frequently reported. This brings into question whether these restraints were not used or they were not measured in these studies.

There were considerably fewer studies reporting "any chemical restraints" in this review ($n = 7$) compared to the number reporting "any physical restraints" ($n = 50$). This may have been a consequence of our inclusion criteria as the intent to restrain needing to be established. Many of our excluded studies had described a focus on the reduction in psychotropic medication use without specifying whether the purpose of the intervention was for restraint or restraint reduction. We feel that exclusion of these studies was still justified however, as there are many potential reasons for prescribing or de-prescribing a psychotropic medication, of which restraint / restraint reduction is just one of them.

It is possible that prevalence rates may have decreased over the time span of included studies due to development of restraint reduction interventions and policies. However, this would assume that interventions and policies that have been developed have both been effective and widely translated into practice. Evaluating the effectiveness of these interventions was beyond the scope of the present review, but we would caution that in the randomised controlled trials we have included in our review, there has been high variability in approaches used to measure restraint, a common focus on physical or chemical restraint rather than both (meaning that restraint could just be transferring from one approach that is being measured to another approach that is not being measured) and little attention paid to the clinometric properties of the measurement approaches employed.

A strength in our review is that we have conducted analyses to identify potential sources of design-related bias in prevalence studies of this nature. It helps to improve the understanding of what sort of methodological approaches are likely to generate higher or lower prevalence estimates. Another key strength of this review was our inclusion of both physical and chemical restraint, as this provides a more comprehensive account of this problem than examining this problem in isolation. A resident who is restrained has lost personal liberties and is at risk of adverse consequences of restraint regardless of the manner of that restraint. Focusing only on one particular type of restraint therefore provides an inadequate picture of the overall problem and may give a biased account of the success of policies and strategies investigated for minimising restraint.

However, there are limitations that need to be considered. There were insufficient studies with data available to build a multiple meta-regression model for the outcome of "any chemical restraints". Thus, it is currently unclear what factors independently contribute to the heterogeneity in chemical restraint prevalence. Nonetheless, this outcome is important in the appraisal of restraint use in residential aged care due to the possibility of transference of restraint approaches from one form to the other i.e. physical to chemical methods and vice versa. Some studies have demonstrated at least a four-fold variability in the prescription of medication classes including antipsychotics in residential aged care facilities that is unlikely to be explained by differences in resident characteristics alone (Jokanovic et al., 2017). There are also studies that found variations in prescription practice when there was a controversy concerning the appropriateness of antipsychotic medications in the management of behavioural symptoms (Lee et al.,

2004; Rochon et al., 2007). In addition, there is a substantial risk that our overall estimate of "any physical" or "any chemical" restraint prevalence and that of the specific mechanism, equipment and medication used to implement restraint may be biased due to the poor methodological quality of studies included in this review. There was an overall poor or unclear sampling frame used across many studies that might have impacted on the representativeness of the target population, and a paucity of information regarding the reliability or validity of the measurement approaches selected in the included studies. For example, studies that used a controlled trial design frequently reported strong data collection methods but limited sampling methods as compared to other study designs which varied greatly. We also did not search grey literature or studies published in languages other than English, this may have a sizeable impact on the results relating to chemical restraints if these studies do exist.

Future research in the prevalence of restraint use will need to work towards a multinational study of restraint prevalence involving large samples and multiple facilities that uses consistent, reliable and valid measurement approaches. These measurement approaches may likely involve direct observations or a combination of multiple measurement approaches due to their higher reported prevalence rates than other methods. There is also the need to employ a more comprehensive approach to defining and measuring restraint that encompasses physical, chemical, environmental and other restraint approaches. This research should ensure a consistent, valid and reliable data collection approach across the residential aged care facilities involved. Countries / regions in future studies should be selected on the basis of having specific differences in restraint reduction policies so that the impact of different policies on restraint use in residential aged care facilities can be understood. Research into culture, ways to study culture transformation to reduce the use of restraints and exploring health economic aspects of restraint use would be important to inform future strategies to minimize restraint use in practice.

6. Conclusion

The variability in prevalence rates of physical restraint could be explained partly by different measurement approaches and geographical regions. Prevalence rates of physical restraint vary widely across geographical regions with comparatively lower levels in North America than other regions, and between measurement approaches with comparatively higher levels using direct observation and a combination of measurement approaches. Further epidemiological research is indicated that uses consistent, valid and reliable measurement approaches across different regions to understand whether variation in physical restraint use is due to cultural differences such as federal restraint policies that could be replicated in other jurisdictions. In addition, more epidemiological studies that clearly identified medication use for restraint purpose are required to determine the factors for the variability in the prevalence of chemical restraint.

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Declaration of Competing Interest

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Supplementary materials

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